



71 THE NUMBER OF STATES THAT SIGNED THE TREATY TO BAN NUCLEAR MISSILE TESTS

Delegates at the United Nations observe the result of their voting on the Comprehensive Nuclear-Test-Ban Treaty—an outright ban on all nuclear explosions.

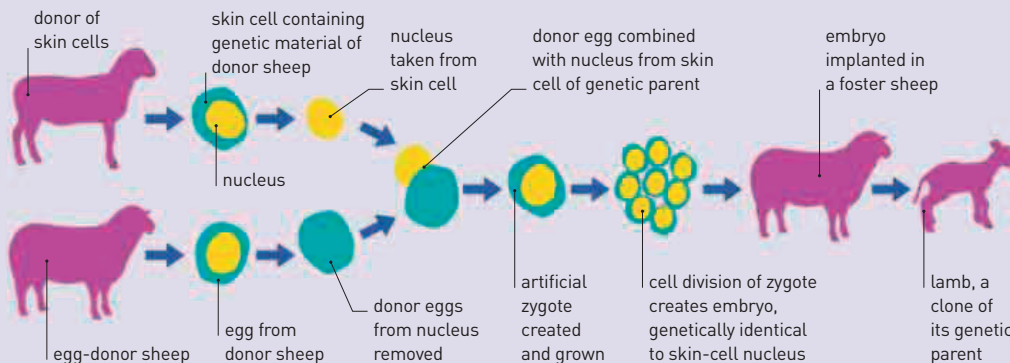
rolled up tightly and cannot be perceived directly. All the theories had inconsistencies, but at a conference at the University of Southern California, American theoretical physicist **Ed Witten** (b.1951) proposed a way of combining them into a **single, super-theory**, which became known as M-theory. To date, it is the most complete “theory of everything,” explaining the existence of the particles in the Standard Model (see 1974), but it is hard to test its validity.

In June 1995, physicists at the University of Colorado **created an unusual state of matter** called a Bose–Einstein condensate (BEC), predicted in the 1920s, in which several particles at a temperature just above absolute zero (see 1847–48) attain exactly the same quantum state, and act as a single system.

In September, physicists at the European Organization for Nuclear Research (CERN), on the border of Switzerland and France, **created antiatoms**—composed of antiprotons and positrons (antielectrons). Antimatter particles are created naturally, in cosmic ray collisions, for example, but when an antiparticle meets a particle, both are annihilated. Modern physics has not explained why matter, rather than antimatter, dominates the Universe.

CLONING

The process that created Dolly the sheep began with the removal of the nucleus (enucleation) of a cell taken from an adult sheep. The nucleus, containing the animal’s DNA, was inserted into an enucleated egg cell from another sheep. The egg was fertilized and then grew into a sheep with a genome identical to that of the original sheep.



In 1995, molecular biologists at the **Institute for Genomic Research** in Maryland completed the first genome sequence of a bacterium.

In 1996, a global collaboration among biologists around the world resulted in the **first complete sequencing of the genome of a eukaryote** (an

“ LIKE AN ICE CREAM CONE, WITH A NEWLY UNCOVERED STAR PLAYING... THE CHERRY ON TOP. ”

Jeff Hester, US physicist, 1995

organism whose cells have a nucleus). Another significant breakthrough in genome science and technology was the **creation of a cloned sheep**, which was **named Dolly**, at the Roslin Institute, Scotland. Clones of many animals, including mammals, had been carried out before, but Dolly was the result of transferring DNA from a cell taken from an adult sheep into an egg—a procedure known as **somatic cell nuclear transfer** (see panel, above).

In September, the UN adopted the Comprehensive Nuclear-Test-Ban Treaty, which places a ban on all nuclear explosions. It has still not been fully ratified. In technology, the **Universal Serial Bus (USB)** connection was launched in 1996, and the first

commercial DVD players became available in Japan. In November 1996, Japanese inventor **Shuji Nakamura** (b.1954) invented a continuous, low-power **blue light LED** (light emitting diode) **laser**. Since blue light has a shorter

wavelength than the red light used in DVD players, Nakamura’s invention allows much more information to be carried on DVD-like disks.

SHUJI NAKAMURA (b.1954)

Born in Ikata, Japan, Shuji Nakamura studied electronic engineering at the University of Tokushima. He made the first practical LEDs (light emitting diodes) to use gallium nitride, resulting in brighter LEDs, and the first LEDs to produce blue light. Nakamura’s development of the blue LED laser represents a milestone in consumer electronics.

